Assessing work	The assessment criteria are shown in the next section. Each criterion scores a maximum of 1 mark. The criteria are not hierarchical. The assessment of the summary of the visit or the case study, together with the planning, the recording of measurements and the analysis is based on documents produced by the students.
For centres marking the written report	The marks for the report should be submitted to Edexcel on the mark sheet for centres provided in the Edexcel *Coursework Guide* that will be available on the Edexcel website www.edexcel.org.uk. Each piece of work should be annotated by the teacher. This can be done by writing the skill code eg A10 near to the appropriate section of the report and ticking the box A10 on the grid below.
For centres not marking the written report	The written report should be submitted to Edexcel as discussed in the Edexcel *Coursework Guide* that will be available on the Edexcel website www.edexcel.org.uk.
Guidance to students	Teachers may provide guidance to students without penalty. Guidance is feedback that a teacher might reasonably be expected to give to a student who asks questions about the work that they are carrying out. In effect, the teacher is being used as a resource. Students may require assistance whereby the teacher needs to tell the student what they have to do. Assistance in this respect carries a penalty. The teacher should record details of any assistance provided on the report.
Important	Students should submit their work for assessment **once** only. Internally assessed work should **not** be given back to students to be improved.

3.3 Assessment criteria

A: Summary of case study or physics-based visit

Ref	Criterion	Mark
S1	Carries out a visit **OR** uses library, consulting a **minimum of three** different sources of information (eg books/websites/journals/magazines/case study provided by Edexcel/ manufacturers' data sheets)	1
S2	States details of visit venue **OR** provides **full** details of sources of information	1
S3	Provides a brief description of the visit **OR** case study	1
S4	Makes correct statement on relevant physics principles	1
S5	Uses relevant specialist terminology correctly	1
S6	Provides one piece of relevant information (eg data, graph, diagram) that is not mentioned in the briefing papers for the visit or case study	1
S7	Briefly discusses context (eg social/environmental/historical)	1
S8	Comments on implication of physics (eg benefits/risks)	1
S9	Explains how the practical relates to the visit or case study	1
	Maximum marks for this section	**9**

B: Planning

Ref	Criterion	Mark
P1	Lists all material required	1
P2	States how to measure one relevant quantity using the most appropriate instrument	1
P3	Explains the choice of the measuring instrument with reference to the scale of the instrument as appropriate and/or the number of measurements to be taken	1
P4	States how to measure a second relevant quantity using the most appropriate instrument	1
P5	Explains the choice of the second measuring instrument with reference to the scale of the instrument as appropriate and/or the number of measurements to be taken	1
P6	Demonstrates knowledge of correct measuring techniques	1
P7	States which is the independent and which is the dependent variable	1
P8	Identifies and states how to control all other relevant variables to make it a fair test	1
P9	Comments on whether repeat readings are appropriate in this case	1
P10	Comments on safety	1
P11	Discusses how the data collected will be used	1
P12	Identifies the main sources of uncertainty and/or systematic error	1
P13	Draws an appropriately labelled diagram of the apparatus to be used	1
P14	Plan is well organised and methodical, using an appropriately sequenced step-by-step procedure	1
	Maximum marks for this section	**14**

C: Implementation and Measurements

Ref	Criterion	Mark
M1	Records all measurements using the correct number of significant figures, tabulating measurements where appropriate	1
M2	Uses correct units **throughout**	1
M3	Obtains an appropriate number of measurements	1
M4	Obtains measurements over an appropriate range	1
	Maximum marks for this section	**4**

D: Analysis

Ref	Criterion	Mark
A1	Produces a graph with appropriately labelled axes and with correct units	1
A2	Produces a graph with sensible scales	1
A3	Plots points accurately	1
A4	Draws line of best fit (either a straight line or a smooth curve)	1
A5	Comments on the trend/pattern obtained	1
A6	Derives relation between two variables or determines constant	1
A7	Discusses/uses related physics principles	1
A8	Attempts to qualitatively consider sources of error	1
A9	Suggests realistic modifications to reduce error/improve experiment	1
A10	Calculates uncertainties	1
A11	Provides a final conclusion	1
	Maximum marks for this section	**11**

E: Report

Ref	Criterion	Mark
R1	Summary contains few grammatical or spelling errors	1
R2	Summary is structured using appropriate subheadings	1
	Maximum marks for this section	**2**

Total marks for this unit	**40**